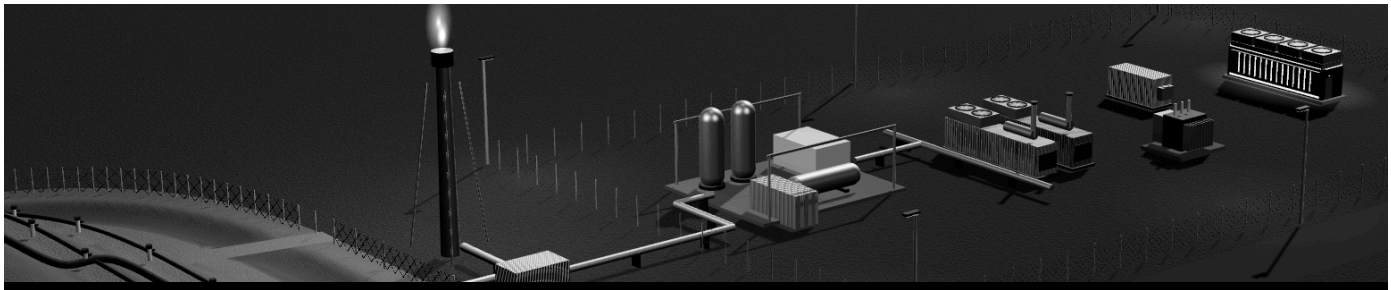
McCain Foods USA, Othello plant wastewater biogas, Othello, Washington: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

1.0 MW	1	6 ac	96k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $72k	$6k to $13k	$6k to $85k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=317&CH4=60

THE MATH

Wastewater biogas sent to the flare (2022, the lower of two reported years)	317 scfm
Methane content (assumed; not published)	60%
Energy in the gas	11.5 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 1.0 MW
Compute (power less cooling and losses, /1.2)	about 0.9 MW
Compute modules (about 1 MW each)	1
Gas engines (running plus one spare)	2 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 6 acres
Gas bought per year (95% run time)	96k MMBtu
Gas payment, up to $0.75/MMBtu	up to $72k
Land lease at $1,000 to $2,000 per acre per year	$6k to $13k

Biogas: methane at about 1,012 Btu per cubic foot. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- 2023 unused gas would give about 1.4 MW
- Boilers keep first call on the gas
- Processing is seasonal: monthly flow matters

Source: EPA Greenhouse Gas Reporting Program (GHGRP), facility 1002287: about 101,000 MMBtu of biogas to the flare in 2022 and about 154,000 MMBtu in 2023. Sized to 2022 (about 317 scfm at an assumed 60% methane). This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.